

VITIM, Russian Federation

WMO: 300540

Lat: 59.4605N

Lon: 112.5670E

Elev: 623

StdP: 14.37

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-53.3	-49.0	-57.1	0.2	-53.6	-52.9	0.2	-48.8	17.5	2.3	15.7	-1.0	0.9	340	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	85.8	66.7	81.9	65.3	78.1	63.7	69.1	80.7	67.1	78.1	65.2	75.3	6.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	95.1	74.2	63.0	88.2	71.9	61.1	82.3	70.2	33.8	80.8	32.1	78.3	30.5	75.6	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.4	13.3	DB	-56.4	92.7	5.8	3.0	-60.5	94.9	-63.9	96.6	-67.1	98.3	-71.3	100.5	(4)
(5)			WB	-56.7	72.3	5.8	2.4	-60.8	74.1	-64.2	75.5	-67.5	76.8	-71.7	78.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	-19.0	-8.8	9.0	29.2	44.5	60.8	65.7	59.9	44.9	26.7	1.0	-17.4		
	DBStd	32.42	18.69	16.21	14.39	8.96	7.47	7.47	5.88	6.29	7.38	10.31	18.11	19.20		
	HDD50	10374	2138	1647	1270	625	206	8	0	6	188	724	1471	2091		
	HDD65	14796	2603	2067	1735	1074	637	165	64	181	604	1189	1921	2556		
	CDD50	1197	0	0	0	0	33	333	485	312	34	0	0	0		
	CDD65	145	0	0	0	0	0	40	84	21	0	0	0	0		
	CDH74	1776	0	0	0	0	26	612	888	245	5	0	0	0		
	CDH80	511	0	0	0	0	4	189	268	50	0	0	0	0		
	WSAvg	5.4	4.3	4.8	5.2	6.0	6.3	5.6	5.4	5.2	5.8	6.0	5.4	4.3		
	PrecAvg	17.3	1.1	0.7	0.6	0.7	1.4	2.0	2.3	2.3	1.9	1.6	1.5	1.3		
	PrecMax	25.5	2.4	1.5	1.5	1.7	3.5	4.8	5.7	6.3	4.2	3.9	2.7	2.3		
(15) Precipitation	PrecMin	10.5	0.3	0.1	0.1	0.0	0.3	0.0	0.0	0.4	0.5	0.4	0.7	0.4		
	PrecStd	3.0	0.4	0.4	0.3	0.3	0.6	1.1	1.4	1.1	0.9	0.7	0.5	0.4		
	DB	23.0	29.9	45.5	58.0	77.4	90.1	91.5	85.9	71.1	51.5	37.6	27.9			
	MCWB	20.9	27.1	36.6	44.5	58.7	67.1	69.7	69.3	57.2	43.3	34.5	26.0			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	18.3	25.3	37.7	52.1	69.8	85.7	86.5	80.0	64.7	46.1	34.1	21.4			
	2% MCWB	17.0	23.2	31.8	41.0	53.1	65.4	67.7	65.2	53.2	40.6	32.2	19.9			
	5% DB	13.6	19.8	33.3	47.3	65.2	81.3	82.7	76.2	60.1	42.2	31.4	16.5			
	5% MCWB	12.5	18.1	29.1	38.1	50.3	63.7	66.3	63.3	51.5	38.2	29.6	15.3			
	10% DB	8.2	14.4	29.5	42.9	60.3	76.9	79.0	72.2	56.1	39.2	27.1	10.8			
	10% MCWB	7.3	13.1	25.8	35.2	48.3	61.9	65.1	61.2	50.0	35.9	25.4	9.7			
	0.4% WB	21.5	27.3	36.6	45.4	59.7	70.8	72.0	70.6	58.6	44.4	35.2	25.8			
	0.4% MCDB	22.8	29.3	44.2	56.5	74.3	85.5	86.3	81.7	68.6	48.9	37.6	27.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	17.0	23.4	32.5	41.7	55.2	67.4	69.9	67.0	55.5	41.6	32.5	20.1			
	2% MCDB	18.3	25.3	37.4	51.1	66.5	80.4	81.8	76.9	61.5	45.6	34.0	21.2			
	5% WB	12.5	18.0	29.1	38.6	51.7	65.2	68.1	64.9	52.9	38.7	29.5	15.3			
	5% MCDB	13.7	19.6	33.1	46.0	63.3	77.9	79.2	73.3	58.6	41.6	31.5	16.4			
	10% WB	7.2	13.1	25.9	36.1	48.7	63.0	66.3	62.8	50.4	36.1	25.3	9.9			
	10% MCDB	8.2	14.5	29.4	42.0	59.3	74.7	76.6	69.9	55.7	38.8	27.1	10.9			
	MDBR	15.2	19.4	25.9	20.9	21.5	22.6	20.3	19.1	14.8	11.9	13.9	13.9			
	5% DB	17.8	17.3	23.1	26.4	32.6	32.1	29.4	28.0	23.4	15.0	11.8	15.6			
(36) Mean Daily Temperature Range	MCWBR	17.1	16.0	18.0	16.5	17.8	14.4	13.0	13.8	13.0	10.3	11.4	15.1			
	MCDBR	17.9	16.9	22.6	23.5	29.4	27.1	24.3	24.3	19.4	13.6	11.5	15.2			
	MCWBR	17.2	15.7	18.2	15.1	17.0	13.0	12.1	12.6	11.6	10.2	11.3	14.8			
	MCDBR	17.2	15.7	18.2	15.1	17.0	13.0	12.1	12.6	11.6	10.2	11.3	14.8			
(40) Clear-Sky Solar Irradiance	taub	0.260	0.296	0.318	0.352	0.381	0.421	0.454	0.417	0.343	0.308	0.263	0.238			
	taud	1.962	1.998	2.045	2.091	2.263	2.274	2.193	2.267	2.445	2.304	2.014	1.922			
	Ebn at Noon	155	216	253	266	268	258	244	245	250	218	153	117			
	Edh at Noon	17	28	36	42	38	39	41	35	25	21	16	12			
(44) All-Sky Solar Radiation	RadAvg	168	461	954	1418	1598	1793	1622	1283	772	359	180	96			
	RadStd	14	41	79	97	126	183	156	106	77	50	27	9			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page